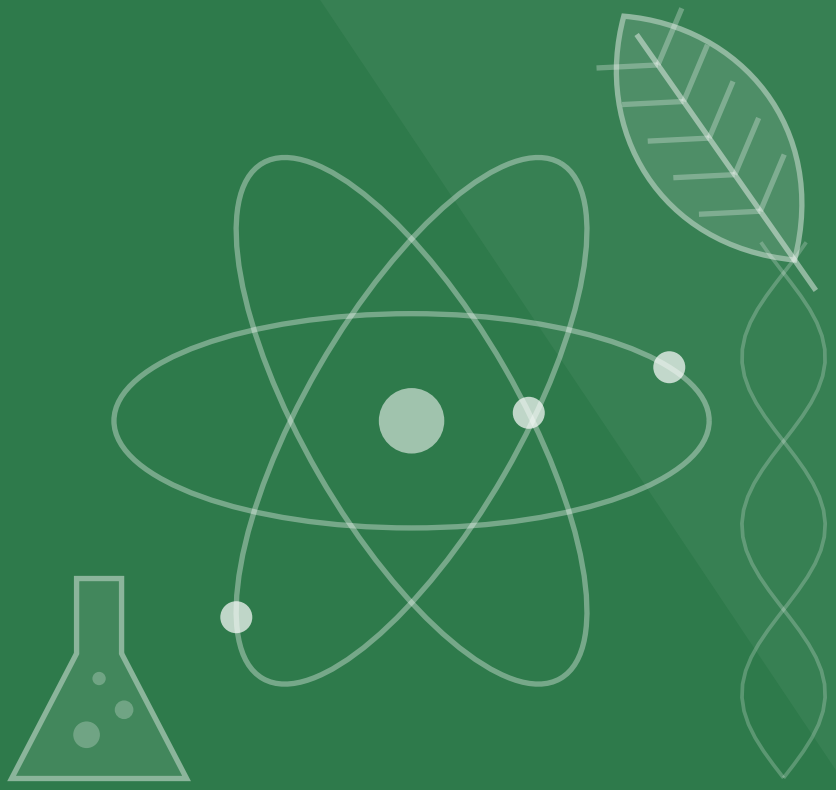
VENVLEARNING

TEACHER EDITION



Science

GRADE 7

Guided lessons, pacing, and complete answer keys

2 units

VENVLEARNING

Grade 7
Science

TEACHER EDITION

Unit 1 Cycles of Matter and Energy

Unit 2 Properties and Changes of Matter

A VenvLearning family homeschool curriculum

How to use this Teacher Edition

This book holds everything needed to teach the course: unit overviews, a plan for every lesson with what to say and do, the student lesson with Try It answers, and answer keys for every workbook page, weekly check and unit test.

- Daily routine (about 45–55 minutes): Engage 5 · Investigate 15 · Explain 10 · Guided Practice 7 · Independent 10 · Exit Ticket 3.
- Say: lines are a suggested script. Use your own words.
- Two lessons a week, then the Weekly Check (about 15 minutes), then corrections or a project.
- Each plan's Pacing row and each unit's pacing guide give the week and lesson order. Set the days to fit your own calendar.

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UNIT 1 · TEACHER OVERVIEW

Cycles of Matter and Energy

Science practices, Earth's spheres, the water, carbon, oxygen, and nitrogen cycles, photosynthesis, respiration, food webs, and biodiversity.

Every atom in your body has been somewhere else before: in the air, the ocean, a plant, or another animal. In this unit you'll learn how scientists design experiments and explain data, then use those skills to trace how matter cycles and energy flows through Earth's systems and living things.

Big ideas

- Identify variables and write CER explanations from data.
- Model the water, carbon, oxygen, and nitrogen cycles.
- Explain photosynthesis and cellular respiration as linked processes.
- Model energy flow in food webs and evaluate ways to protect biodiversity.

Standards

Code	Standard
MS-LS1-6, 1-7	Photosynthesis; cellular respiration.
MS-LS2-3	Cycling of matter and flow of energy in ecosystems.
MS-LS2-5	Maintaining biodiversity and ecosystem services.
MS-ESS2-4	The water cycle driven by the sun and gravity.
SEP 3, 4, 6	Investigations; data analysis; explanations.

What students should already know

- Plants need light, water, and air.
- Food chains (Grade 5).

Materials for the unit

- Antacid tablets, cups, thermometer, stopwatch

- Zip bags, food coloring
- Spinach, syringe, baking soda, lamp
- Yeast, sugar, bottles, balloons
- String and sticks for a survey square

Pacing guide

Teach the lessons in order, then give the week's check. Set the days to fit your own calendar.

Week	Order	Lesson	Title
Week 1	Lesson 1 of 2	1.1	Variables, Data, and Scientific Explanations
Week 1	Lesson 2 of 2	1.2	Earth's Spheres and the Water Cycle
Week 1	End of week	Check	Weekly Check 1: Variables and the Water Cycle
Week 2	Lesson 1 of 2	1.3	The Carbon and Oxygen Cycles
Week 2	Lesson 2 of 2	1.4	The Nitrogen Cycle
Week 2	End of week	Check	Weekly Check 2: Carbon, Oxygen, and Nitrogen
Week 3	Lesson 1 of 2	1.5	Photosynthesis
Week 3	Lesson 2 of 2	1.6	Cellular Respiration
Week 3	End of week	Check	Weekly Check 3: Photosynthesis and Respiration
Week 4	Lesson 1 of 2	1.7	Food Webs and Energy Pyramids
Week 4	Lesson 2 of 2	1.8	Biodiversity and Ecosystem Services
Week 4	Before the test	Review	Unit 1 Review
Week 4	End of unit	Test	Unit 1 Test

TEACHING THIS UNIT AT HOME

Bake bread together and talk about how yeast respiration makes it rise.

UNIT VOCABULARY

independent variable	The variable that is changed.
dependent variable	The variable that is measured.
photosynthesis	Light + CO_2 + water → glucose + oxygen.
cellular respiration	Glucose + oxygen → CO_2 + water + energy.
nitrogen fixation	Changing N_2 into usable forms.
biodiversity	The variety of life in an area.

LESSON 1.1 · TEACHER PLAN
Week 1 · Lesson 1 of 2

Unit 1: Cycles of Matter and Energy

Lesson 1.1: Variables, Data, and Scientific Explanations

Student goal	I can identify independent, dependent, and controlled variables and write a scientific explanation using CER.
Objective	Students identify variables and write CER explanations.
Standard	SEP 3, 4, 6 — Plan investigations; analyze data; construct explanations.
Pacing	Unit 1, week 1, lesson 1 of 2
Time	About 50 minutes
Materials	Tablets or sugar, cups, thermometer, stopwatch, workbook Lesson 1.1
Get ready	Read the student lesson, gather and test the investigation materials, and answer the Check Your Understanding questions.

Lesson Flow

Part	Min	What to do and say
Engage	5	Fizzing question.
Investigate	15	Temperature and Dissolving.
Explain	10	Teach variable types and the If–then–because hypothesis. Model CER.
Guided Practice	7	Check Your Understanding 1–3.
Independent	10	Workbook lesson pages: Warm-Up through Exit Ticket.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Confusing independent and dependent variables.

SUPPORT

- Mnemonic: I change the Independent; the Dependent Depends on it.

EXTEND

- Design a second experiment testing stirring speed.

STUDENT LESSON 1.1 · WITH ANSWERS

Lesson 1.1: Variables, Data, and Scientific Explanations

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 1.1 in this book with your teacher.
- [ ] Do the Investigate activity: Temperature and Dissolving.
- [ ] Answer the Check Your Understanding questions in Workbook Lesson 1.1, Part A.
- [ ] Complete the rest of Workbook Lesson 1.1 (Parts B–E).

I WONDER

Does a fizzing tablet dissolve faster in hot water or cold water? How could you find out fairly?

A controlled experiment tests how one factor affects another. The independent variable is what you change. The dependent variable is what you measure. Everything else stays the same as controlled variables. A good hypothesis is written so it can be tested: "If..., then..., because...."

Part	Fizzing-tablet experiment
question	How does water temperature affect how long a tablet takes to dissolve?
hypothesis	If the water is warmer, then the tablet will dissolve faster, because particles move faster at higher temperatures.
independent variable	water temperature
dependent variable	time to dissolve (seconds)
controlled variables	amount of water, type and size of tablet, cup, no stirring

INVESTIGATE · TEMPERATURE AND DISSOLVING

How does water temperature affect dissolving time?

You need: 3 identical effervescent antacid tablets (or 3 equal spoonfuls of sugar); 3 clear cups; cold, room-temperature, and warm tap water; thermometer; stopwatch.

1. Measure 200 mL of water into each cup and record each temperature.
2. Drop one tablet into each cup at the same moment and start the timer.
3. Record the time when each tablet is completely dissolved.
4. Repeat if you can, and average the trials.

Record: a data table and a line or bar graph with temperature on the x-axis and time on the y-axis.

Safety: Use warm tap water, not boiling water. Do not swallow the tablets or solutions.

EXPLAIN IT · WRITE A SCIENTIFIC EXPLANATION

Data: 8°C → 95 s; 21°C → 52 s; 40°C → 24 s

1 Claim: Tablets dissolve faster in warmer water.

2 Evidence: dissolving time dropped from 95 s at 8°C to 52 s at 21°C to 24 s at 40°C.

3 Reasoning: particles in warmer water move faster, so they collide with the tablet more often and break it apart sooner.

Explanation: A complete explanation connects the evidence to a scientific idea.

★ KEY IDEA

Change one independent variable, measure the dependent variable, control everything else, and explain results with CER.

✎ CHECK YOUR UNDERSTANDING

Answers are shown in color.

1. Plant height is measured after different amounts of fertilizer. Independent variable?

Answer: amount of fertilizer

2. Same experiment: dependent variable?

Answer: plant height

3. Name two controlled variables.

Answer: e.g. type of plant, water, light, soil, pot size

Workbook Answer Key · Lesson 1.1

#	Question	Answer
1	Plant height is measured after different amounts of fertilizer. Independent variable?	amount of fertilizer
2	Same experiment: dependent variable?	plant height
3	Name two controlled variables.	e.g. type of plant, water, light, soil, pot size
4	The variable you change	independent variable
5	The variable you measure	dependent variable

#	Question	Answer
6	Does the height of a ramp affect how far a car rolls? Independent:	ramp height
7	Same: dependent:	distance the car rolls
8	Same: two controlled variables:	same car, same surface, same release method
9	Write a hypothesis for the ramp experiment.	If the ramp is higher, then the car will roll farther, because it has more energy at the bottom.
10	Data: 0 g salt → water boils at 100.0°C; 20 g → 100.3°C; 40 g → 100.7°C. Write a CER.	Claim: adding salt raises boiling point. Evidence: 100.0 → 100.3 → 100.7°C as salt increases. Reasoning: dissolved particles change how water molecules escape.
11	Why must controlled variables stay the same?	So any change in the dependent variable can be linked to the independent variable alone.

LESSON 1.2 · TEACHER PLAN
Week 1 · Lesson 2 of 2
Unit 1: Cycles of Matter and Energy

Lesson 1.2: Earth's Spheres and the Water Cycle

Student goal	I can describe Earth's four spheres and model how energy from the sun and gravity drive the water cycle.
Objective	Students model the water cycle within Earth's spheres.
Standard	MS-ESS2-4 — Develop a model of the cycling of water driven by the sun and gravity.
Pacing	Unit 1, week 1, lesson 2 of 2
Time	About 50 minutes
Materials	Zip bags, water, food coloring, tape, workbook Lesson 1.2
Get ready	Read the student lesson, gather and test the investigation materials, and answer the Check Your Understanding questions.

Lesson Flow

Part	Min	What to do and say
Engage	5	Dinosaur water.
Investigate	15	Water Cycle in a Bag (set up today, observe all week).
Explain	10	Teach the four spheres; the cycle diagram.
Guided Practice	7	Check Your Understanding 1–3.
Independent	10	Workbook. The Weekly Check comes next.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Thinking water is “used up.”

SUPPORT

- Sphere sorting cards.

EXTEND

- Research how much of Earth's water is fresh and available.

STUDENT LESSON 1.2 · WITH ANSWERS

Lesson 1.2: Earth's Spheres and the Water Cycle

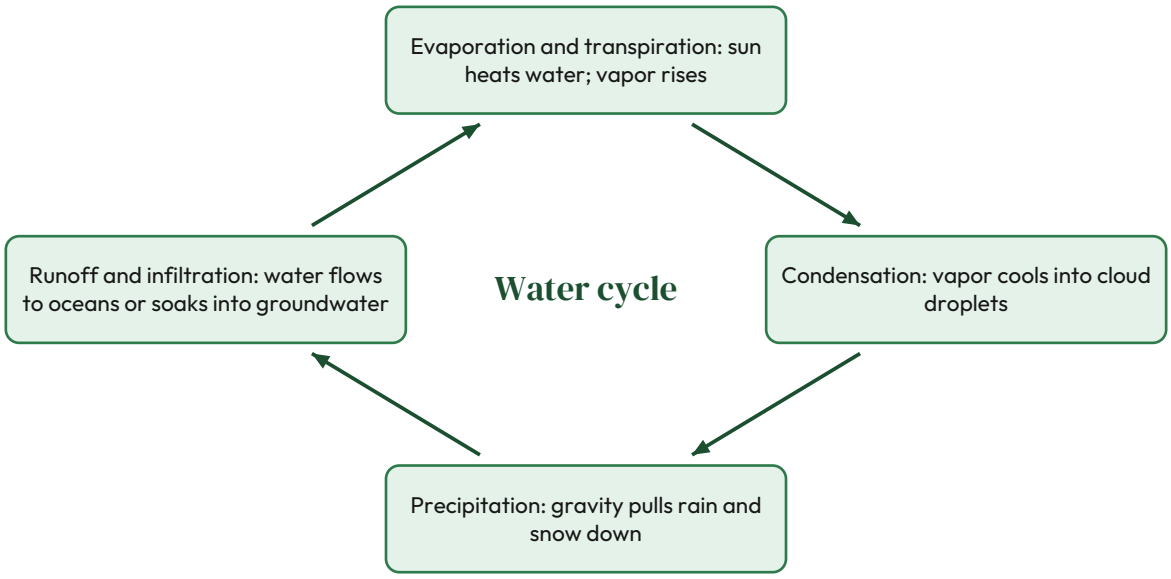
STUDENT'S ASSIGNMENT

- [ ] Read Lesson 1.2 in this book with your teacher.
- [ ] Do the Investigate activity: Water Cycle in a Bag.
- [ ] Answer the Check Your Understanding questions in Workbook Lesson 1.2, Part A.
- [ ] Complete the rest of Workbook Lesson 1.2 (Parts B–D).

I WONDER

The water you drank today may once have been in a dinosaur, a cloud, or an ocean. How could that be?

Scientists study Earth as a system of four interacting parts, or spheres: the geosphere, hydrosphere, atmosphere, and biosphere. Matter, like water, constantly moves among them. Earth has about the same amount of water today as it had long ago; it just keeps cycling.



Two forces drive the cycle. Energy from the sun evaporates water and powers winds. Gravity pulls precipitation down and makes water flow downhill into rivers, lakes, and oceans, and soak into the ground.

INVESTIGATE · WATER CYCLE IN A BAG

Can you model the water cycle?

You need: a zip-top bag; 1/4 cup water with a few drops of blue food coloring; tape; a sunny window; marker.

1. Draw a sun, cloud, and waves on the bag.
2. Pour in the water and seal the bag.
3. Tape it to a sunny window.
4. Observe it after 1 hour, after 1 day, and after 2 days.

Record: observations and a labeled sketch showing evaporation, condensation, and precipitation in the bag.

★ KEY IDEA

Water cycles among Earth's spheres, driven by energy from the sun and by gravity.

✎ CHECK YOUR UNDERSTANDING

Answers are shown in color.

1. Which sphere includes clouds and air?

Answer: atmosphere

2. What process returns water vapor from plants?

Answer: transpiration

3. What force makes rain fall and rivers flow?

Answer: gravity

Workbook Answer Key · Lesson 1.2

#	Question	Answer
1	Which sphere includes clouds and air?	atmosphere
2	What process returns water vapor from plants?	transpiration
3	What force makes rain fall and rivers flow?	gravity
4	All living things make up the...	biosphere
5	Water vapor cooling into droplets is...	condensation
6	Which sphere does a river belong to?	hydrosphere
7	Which sphere does a mountain belong to?	geosphere
8	What provides the energy for evaporation?	the sun
9	What happens to water that soaks into soil?	it becomes groundwater (infiltration)

#	Question	Answer
10	Describe one interaction between two spheres in the water cycle.	e.g. water evaporates from the ocean (hydrosphere) into the atmosphere.
11	Explain the droplets on the inside of the bag using the words evaporation and condensation.	Sunlight evaporated water; the vapor touched the cooler bag and condensed into droplets.
12	Name the two things that drive the water cycle and what each does.	Sun's energy causes evaporation; gravity causes precipitation and runoff.

WEEKLY CHECK · ANSWER KEY
After Lesson 1.2

About 15 minutes

Weekly Check 1: Variables and the Water Cycle — Answer Key

Answers

#	Question	Answer
1	Does the amount of sunlight affect bean plant growth? Independent variable:	amount of sunlight
2	Dependent variable:	plant growth (height)
3	One controlled variable:	e.g. water, soil, type of bean
4	Water vapor cooling into droplets is…	condensation
5	Which sphere includes all living things?	biosphere
6	Name the two drivers of the water cycle and what each does.	Sun: evaporation; gravity: precipitation and runoff.

LESSON 1.3 · TEACHER PLAN
Week 2 · Lesson 1 of 2
Unit 1: Cycles of Matter and Energy

Lesson 1.3: The Carbon and Oxygen Cycles

Student goal	I can explain how carbon and oxygen cycle among living things, the atmosphere, the oceans, and rocks.
Objective	Students trace the carbon and oxygen cycles.
Standard	MS-LS2-3 — Develop a model to describe the cycling of matter among living and nonliving parts.
Pacing	Unit 1, week 2, lesson 1 of 2
Time	About 50 minutes
Materials	Station cards, dice, workbook Lesson 1.3
Get ready	Read the student lesson, gather and test the investigation materials, and answer the Check Your Understanding questions.

Lesson Flow

Part	Min	What to do and say
Engage	5	Carbon everywhere.
Investigate	15	Follow a Carbon Atom.
Explain	10	Teach the process table.
Guided Practice	7	Check Your Understanding 1–3.
Independent	10	Workbook lesson pages: Warm-Up through Exit Ticket.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Thinking plants don't respire.

SUPPORT

- Two-column in/out chart.

EXTEND

- Graph a year of carbon dioxide data and explain the seasonal pattern.

STUDENT LESSON 1.3 · WITH ANSWERS

Lesson 1.3: The Carbon and Oxygen Cycles

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 1.3 in this book with your teacher.
- [ ] Do the Investigate activity: Follow a Carbon Atom.
- [ ] Answer the Check Your Understanding questions in Workbook Lesson 1.3, Part A.
- [ ] Complete the rest of Workbook Lesson 1.3 (Parts B–D).

I WONDER

Carbon is in your body, in a diamond, in soda bubbles, and in the air. How can one element be in all of these?

Carbon is the building block of living things. Carbon atoms are never used up; they cycle. Plants take carbon dioxide from the air during photosynthesis and build sugars. Animals eat plants. Living things release carbon dioxide during cellular respiration, and decomposers release it as they break down dead matter. Oceans absorb and release carbon dioxide too. Over millions of years, some carbon becomes buried as fossil fuels; burning them releases that carbon quickly.

The oxygen cycle is linked to the carbon cycle: photosynthesis releases oxygen, and respiration and combustion use oxygen. Today air is about 21 percent oxygen and about 0.04 percent carbon dioxide.

Process	Carbon dioxide	Oxygen
photosynthesis	taken in	released
cellular respiration	released	taken in
decomposition	released	taken in
combustion (burning fuels)	released	taken in
ocean absorption	taken in (dissolves in water)	—

INVESTIGATE · FOLLOW A CARBON ATOM

What paths can one carbon atom take?

You need: station cards: Atmosphere, Plant, Animal, Decomposer, Ocean, Fossil Fuel; one die.

1 Start at the Atmosphere station.

2 Roll the die; each station card tells where a roll sends you and why (e.g. 1–3: a plant takes you in during photosynthesis).

3 Record each move for 10 rolls.

4 Compare paths with a partner.

Record: your atom's path as a flow chart, with the process that moved it at each arrow.

★ KEY IDEA

Photosynthesis removes carbon dioxide and adds oxygen; respiration, decomposition, and combustion do the opposite.

✎ CHECK YOUR UNDERSTANDING

Answers are shown in color.

1. Which process removes carbon dioxide from the air?

Answer: photosynthesis

2. Name two processes that release carbon dioxide.

Answer: respiration, decomposition, or combustion

3. About what percent of air is oxygen?

Answer: 21 percent

Workbook Answer Key · Lesson 1.3

#	Question	Answer
1	Which process removes carbon dioxide from the air?	photosynthesis
2	Name two processes that release carbon dioxide.	respiration, decomposition, or combustion
3	About what percent of air is oxygen?	21 percent
4	Coal, oil, and natural gas are...	fossil fuels
5	Organisms that break down dead matter	decomposers
6	A tree takes in CO_2. Process?	photosynthesis
7	A car burns gasoline. Process?	combustion
8	A fungus breaks down a log. Process?	decomposition
9	You exhale. Process?	cellular respiration

#	Question	Answer
10	Explain why burning fossil fuels adds carbon to the atmosphere faster than natural processes remove it.	Burning releases carbon stored over millions of years in a short time; photosynthesis and oceans can't absorb it all as fast.
11	How are the carbon and oxygen cycles connected?	Processes that take in CO_2 release O_2 (photosynthesis) and processes that release CO_2 use O_2 (respiration, combustion).

LESSON 1.4 · TEACHER PLAN
Week 2 · Lesson 2 of 2
Unit 1: Cycles of Matter and Energy

Lesson 1.4: The Nitrogen Cycle

Student goal	I can explain how nitrogen moves from the air into living things and back.
Objective	Students model the nitrogen cycle.
Standard	MS-LS2-3 — Develop a model to describe the cycling of matter.
Pacing	Unit 1, week 2, lesson 2 of 2
Time	About 50 minutes
Materials	Bean or pea plant, magnifier, workbook Lesson 1.4
Get ready	Read the student lesson, gather and test the investigation materials, and answer the Check Your Understanding questions.

Lesson Flow

Part	Min	What to do and say
Engage	5	Why not breathe it?
Investigate	15	Root Nodule Search.
Explain	10	Teach the cycle; Carver and crop rotation.
Guided Practice	7	Check Your Understanding 1–3.
Independent	10	Workbook. The Weekly Check comes next.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Thinking plants take N_2 from the air.

SUPPORT

- Cycle with picture cards.

EXTEND

- Research nitrogen runoff and the Gulf of Mexico dead zone.

STUDENT LESSON 1.4 · WITH ANSWERS

Lesson 1.4: The Nitrogen Cycle

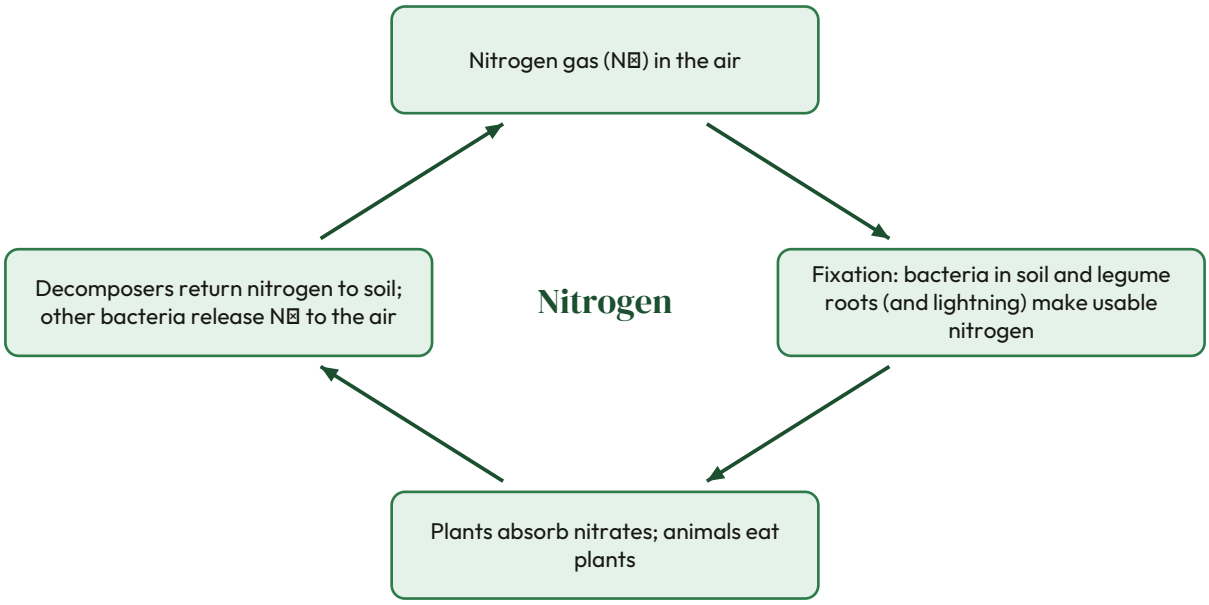
STUDENT'S ASSIGNMENT

- [ ] Read Lesson 1.4 in this book with your teacher.
- [ ] Do the Investigate activity: Root Nodule Search.
- [ ] Answer the Check Your Understanding questions in Workbook Lesson 1.4, Part A.
- [ ] Complete the rest of Workbook Lesson 1.4 (Parts B–D).

I WONDER

Air is about 78 percent nitrogen, and your body needs nitrogen to build proteins. Why can't you just breathe it in and use it?

Living things need nitrogen to build proteins and DNA, but most organisms can't use nitrogen gas (N_2) directly. It has to be fixed into compounds such as ammonia and nitrates.



Farmers use this cycle. Planting legumes such as soybeans or peanuts adds usable nitrogen to the soil. Peanut farmer and scientist George Washington Carver taught Southern farmers to rotate cotton with peanuts and other legumes to restore worn-out soil.

INVESTIGATE · ROOT NODULE SEARCH

Where do nitrogen-fixing bacteria live?

You need: a bean or pea plant (or photos from your teacher); magnifying glass; water.

1 Gently dig up the plant and rinse the soil from its roots.

2 Look for small round bumps (nodules) on the roots.

3 Sketch the roots and nodules.

Record: a labeled sketch and an explanation of what the bacteria in the nodules do.

★ KEY IDEA

Bacteria fix nitrogen gas into usable forms; plants absorb it, animals eat plants, and decomposers and bacteria return it.

✎ CHECK YOUR UNDERSTANDING

Answers are shown in color.

1. What percent of air is nitrogen?

Answer: about 78 percent

2. What does nitrogen fixation do?

Answer: changes N_2 into forms plants can use

3. Why do farmers plant legumes?

Answer: Their root bacteria add usable nitrogen to the soil.

Workbook Answer Key · Lesson 1.4

#	Question	Answer
1	What percent of air is nitrogen?	about 78 percent
2	What does nitrogen fixation do?	changes N_2 into forms plants can use
3	Why do farmers plant legumes?	Their root bacteria add usable nitrogen to the soil.
4	Living things use nitrogen to build...	proteins (and DNA)
5	A bean plant is a...	legume
6	Can animals use nitrogen gas directly?	no
7	How do animals get nitrogen?	by eating plants or other animals
8	Name a non-living way nitrogen is fixed.	lightning
9	What returns nitrogen from dead organisms to soil?	decomposers
10	Explain why rotating crops with legumes can help soil.	Legumes host bacteria that fix nitrogen, restoring nitrogen other crops used up.

#	Question	Answer
11	Trace one nitrogen atom from the air into your body.	Air → bacteria fix it → plant absorbs nitrate → I eat the plant (or an animal that ate it).

WEEKLY CHECK · ANSWER KEY
After Lesson 1.4

About 15 minutes

Weekly Check 2: Carbon, Oxygen, and Nitrogen — Answer Key

Answers

#	Question	Answer
1	Which process removes CO_2 from air?	photosynthesis
2	Name a process that releases CO_2.	respiration, decomposition, or combustion
3	About what percent of air is nitrogen?	78 percent
4	What do nitrogen-fixing bacteria do?	change N_2 into usable forms
5	Why does burning fossil fuels change the carbon cycle?	It releases stored carbon quickly, adding CO_2 faster than it is removed.

LESSON 1.5 · TEACHER PLAN
Week 3 · Lesson 1 of 2
Unit 1: Cycles of Matter and Energy

Lesson 1.5: Photosynthesis

Student goal	I can explain how plants use light energy to make sugar and oxygen from carbon dioxide and water.
Objective	Students explain photosynthesis and test light's effect.
Standard	MS-LS1-6 — Construct a scientific explanation for the role of photosynthesis in the cycling of matter and flow of energy.
Pacing	Unit 1, week 3, lesson 1 of 2
Time	About 50 minutes
Materials	Spinach, syringe, baking soda, soap, cups, lamp, workbook Lesson 1.5
Get ready	Read the student lesson, gather and test the investigation materials, and answer the Check Your Understanding questions.

Lesson Flow

Part	Min	What to do and say
Engage	5	Oak tree mass.
Investigate	15	Floating Leaf Disks.
Explain	10	Teach the equation; energy transformation.
Guided Practice	7	Check Your Understanding 1–3.
Independent	10	Workbook lesson pages: Warm-Up through Exit Ticket.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Believing plants get their mass from soil.

SUPPORT

- Color-coded equation.

EXTEND

- Test different colors of light with cellophane filters.

STUDENT LESSON 1.5 · WITH ANSWERS

Lesson 1.5: Photosynthesis

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 1.5 in this book with your teacher.
- [ ] Do the Investigate activity: Floating Leaf Disks.
- [ ] Answer the Check Your Understanding questions in Workbook Lesson 1.5, Part A.
- [ ] Complete the rest of Workbook Lesson 1.5 (Parts B–D).

I WONDER

A huge oak tree grows from a tiny acorn. Where does all that mass come from?

Surprisingly, most of a tree's mass comes from the air! In photosynthesis, chloroplasts in leaf cells use light energy captured by chlorophyll to combine carbon dioxide from the air with water from the soil. They produce glucose, which the plant uses for energy and to build its body, and release oxygen.

Word equation	carbon dioxide + water —(light energy)→ glucose + oxygen
Chemical equation	$6CO_2 + 6H_2O \xrightarrow{\text{light}} C_6H_{12}O_6 + 6O_2$

Photosynthesis changes light energy into chemical energy stored in sugar. Almost every food chain on Earth begins with this process.

INVESTIGATE · FLOATING LEAF DISKS

Does light intensity affect the rate of photosynthesis?

You need: fresh spinach leaves; a straw or hole punch; a plastic syringe (no needle); baking soda solution (1/8 tsp in 300 mL water) with 1 drop dish soap; 2 clear cups; a lamp.

1. Punch 20 disks from the leaves.
2. Put 10 disks in the syringe with solution, push out the air, cover the tip, and pull the plunger to remove gas from the leaves until the disks sink. Repeat for the other 10.
3. Put 10 sunk disks in a cup under the lamp and 10 in a cup in a dark place.
4. Count floating disks every minute for 15 minutes.

Record: a line graph of floating disks vs time for both cups, and a CER explaining the difference (oxygen produced in photosynthesis makes disks float).

★ KEY IDEA

Photosynthesis: carbon dioxide + water + light energy → glucose + oxygen, in chloroplasts.

✎ CHECK YOUR UNDERSTANDING

Answers are shown in color.

1. Where does most of a tree's mass come from?

Answer: carbon dioxide from the air

2. What are the products of photosynthesis?

Answer: glucose and oxygen

3. What pigment captures light?

Answer: chlorophyll

Workbook Answer Key · Lesson 1.5

#	Question	Answer
1	Where does most of a tree's mass come from?	carbon dioxide from the air
2	What are the products of photosynthesis?	glucose and oxygen
3	What pigment captures light?	chlorophyll
4	Reactants of photosynthesis	carbon dioxide and water
5	Organelle where it happens	chloroplast
6	Disks floating after 10 min: lamp cup 9; dark cup 0. Which had more photosynthesis?	the lamp cup
7	What gas made the disks float?	oxygen
8	What was the independent variable?	light (lamp vs dark)
9	Write a CER for the leaf disk results.	Claim: light increases photosynthesis. Evidence: 9 vs 0 disks floating. Reasoning: photosynthesis needs light and makes oxygen, which fills the leaf spaces and makes disks float.
10	Explain how photosynthesis transforms energy.	It changes light energy into chemical energy stored in glucose.

LESSON 1.6 · TEACHER PLAN
Week 3 · Lesson 2 of 2
Unit 1: Cycles of Matter and Energy

Lesson 1.6: Cellular Respiration

Student goal	I can explain how cells release energy from food through cellular respiration.
Objective	Students explain cellular respiration.
Standard	MS-LS1-7 — Develop a model of how food is rearranged through chemical reactions to release energy.
Pacing	Unit 1, week 3, lesson 2 of 2
Time	About 50 minutes
Materials	Bottles, balloons, yeast, sugar, warm water, string, ruler, workbook Lesson 1.6
Get ready	Read the student lesson, gather and test the investigation materials, and answer the Check Your Understanding questions.

Lesson Flow

Part	Min	What to do and say
Engage	5	Breakfast power.
Investigate	15	Yeast Balloons.
Explain	10	Teach the equation; compare with photosynthesis.
Guided Practice	7	Check Your Understanding 1–3.
Independent	10	Workbook. The Weekly Check comes next.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Confusing respiration with breathing.

SUPPORT

- Side-by-side equation organizer.

EXTEND

- Bake bread and explain the rise.

STUDENT LESSON 1.6 · WITH ANSWERS

Lesson 1.6: Cellular Respiration

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 1.6 in this book with your teacher.
- [ ] Do the Investigate activity: Yeast Balloons.
- [ ] Answer the Check Your Understanding questions in Workbook Lesson 1.6, Part A.
- [ ] Complete the rest of Workbook Lesson 1.6 (Parts B–D).

I WONDER

You ate breakfast hours ago. How is that food powering your brain right now?

All organisms, including plants, need energy. In cellular respiration, mitochondria in cells break down glucose with oxygen, releasing energy the cell can use. Carbon dioxide and water are given off as waste.

Word equation	glucose + oxygen → carbon dioxide + water + energy
Chemical equation	$C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$ + energy

Notice that respiration is roughly the reverse of photosynthesis. The products of one are the reactants of the other, so matter cycles between plants and animals while energy flows from the sun, through living things, and is eventually lost as heat.

Some organisms, like yeast, can also release energy without oxygen through fermentation, giving off carbon dioxide. That's why bread dough rises.

INVESTIGATE · YEAST BALLOONS

Does yeast need sugar to release carbon dioxide?

You need: 2 empty water bottles; 2 balloons; 2 packets of dry yeast; warm water; 2 tablespoons sugar.

1. Add 1 cup warm water and 1 packet yeast to each bottle.
2. Add sugar to bottle A only. Swirl both.
3. Stretch a balloon over each bottle's mouth.
4. Measure each balloon's circumference every 10 minutes for 30 minutes.

Record: a data table and a graph; explain what filled the balloon and why one grew more.

★ KEY IDEA

Respiration: glucose + oxygen → carbon dioxide + water + energy, in mitochondria. It is the reverse of photosynthesis.

✎ CHECK YOUR UNDERSTANDING

Answers are shown in color.

1. Where does cellular respiration happen?

Answer: in mitochondria

2. What gas filled the balloon?

Answer: carbon dioxide

3. Do plants carry out respiration?

Answer: Yes; all living cells do.

Workbook Answer Key · Lesson 1.6

#	Question	Answer
1	Where does cellular respiration happen?	in mitochondria
2	What gas filled the balloon?	carbon dioxide
3	Do plants carry out respiration?	Yes; all living cells do.
4	Products of respiration	carbon dioxide, water, energy
5	Reactants of respiration	glucose and oxygen
6	Which process stores energy in glucose?	photosynthesis
7	Which process releases energy from glucose?	cellular respiration
8	Which produces oxygen?	photosynthesis
9	Which produces carbon dioxide?	cellular respiration
10	Balloon A (sugar) grew to 28 cm; balloon B (no sugar) to 13 cm. Explain using CER.	Claim: yeast needs sugar to release much CO_2. Evidence: 28 vs 13 cm. Reasoning: yeast breaks down sugar for energy and gives off CO_2.
11	Explain how matter cycles but energy flows between photosynthesis and respiration.	The same atoms move back and forth between the processes; energy enters as light and leaves as heat.

WEEKLY CHECK · ANSWER KEY
After Lesson 1.6

About 15 minutes

Weekly Check 3: Photosynthesis and Respiration — Answer Key

Answers

#	Question	Answer
1	Products of photosynthesis	glucose and oxygen
2	Products of cellular respiration	carbon dioxide, water, energy
3	Organelle for photosynthesis	chloroplast
4	Organelle for respiration	mitochondria
5	Explain how photosynthesis and respiration cycle matter.	Products of each are reactants of the other; atoms move back and forth.

LESSON 1.7 · TEACHER PLAN
Week 4 · Lesson 1 of 2
Unit 1: Cycles of Matter and Energy

Lesson 1.7: Food Webs and Energy Pyramids

Student goal	I can model how energy flows through a food web and explain why less energy reaches each higher level.
Objective	Students model energy flow.
Standard	MS-LS2-3 — Develop a model of cycling of matter and flow of energy among living and nonliving parts.
Pacing	Unit 1, week 4, lesson 1 of 2
Time	About 50 minutes
Materials	Organism cards, yarn, workbook Lesson 1.7
Get ready	Read the student lesson, gather and test the investigation materials, and answer the Check Your Understanding questions.

Lesson Flow

Part	Min	What to do and say
Engage	5	Grasshoppers vs hawks.
Investigate	15	Build a Food Web.
Explain	10	Teach levels; 10 percent rule; pyramid.
Guided Practice	7	Check Your Understanding 1–3.
Independent	10	Workbook lesson pages: Warm-Up through Exit Ticket.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Drawing arrows from eater to eaten.

SUPPORT

- "Gives energy to" arrow phrase.

EXTEND

- Research a Mississippi food web (Gulf Coast marsh or pine forest).

STUDENT LESSON 1.7 · WITH ANSWERS

Lesson 1.7: Food Webs and Energy Pyramids

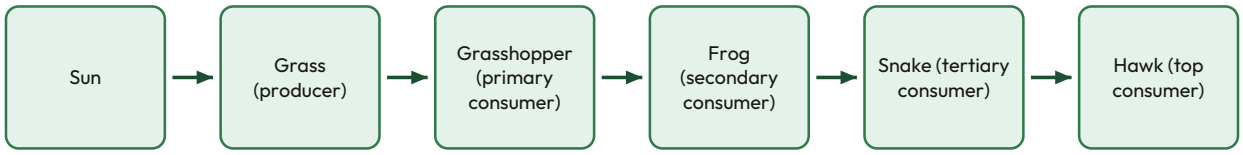
STUDENT'S ASSIGNMENT

- [ ] Read Lesson 1.7 in this book with your teacher.
- [ ] Do the Investigate activity: Build a Food Web.
- [ ] Answer the Check Your Understanding questions in Workbook Lesson 1.7, Part A.
- [ ] Complete the rest of Workbook Lesson 1.7 (Parts B–D).

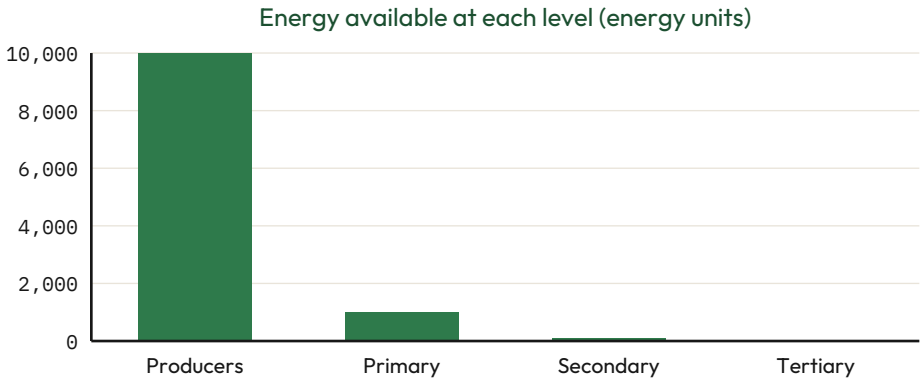
I WONDER

Why are there far more grasshoppers than hawks in a field?

Energy enters most ecosystems as sunlight captured by producers. Primary consumers (herbivores) eat producers, secondary consumers eat herbivores, and so on. Decomposers break down dead organisms at every level. Arrows in a food chain point in the direction energy flows: from the eaten to the eater.



At each level, organisms use most of their energy to live, move, and stay warm, and much is lost as heat. On average only about 10 percent of the energy at one level is stored in the next. That's why an energy pyramid narrows at the top and why there are fewer top predators.



INVESTIGATE · BUILD A FOOD WEB

How are organisms in a forest ecosystem connected?

You need: cards: oak tree, grass, acorn weevil, grasshopper, mouse, rabbit, deer, frog, snake, hawk, fox, mushroom (decomposer); yarn or a pencil.

1 Arrange the cards on a large sheet of paper.

2 Draw arrows from each food to the organism that eats it.

3 Label producers, consumers, and the decomposer.

4 Remove the mouse. Predict two effects on the web.

Record: your food web and your predictions.

★ KEY IDEA

Energy flows from producers to consumers; only about 10 percent passes to each higher level, while matter is recycled by decomposers.

✎ CHECK YOUR UNDERSTANDING

Answers are shown in color.

1. Which way do food-chain arrows point?

Answer: from the organism eaten to the organism that eats it (direction of energy flow)

2. If grass has 20,000 units, about how much reaches grasshoppers?

Answer: 2,000 units

3. Why are top predators few?

Answer: Little energy reaches the top of the pyramid.

Workbook Answer Key · Lesson 1.7

#	Question	Answer
1	Which way do food-chain arrows point?	from the organism eaten to the organism that eats it (direction of energy flow)
2	If grass has 20,000 units, about how much reaches grasshoppers?	2,000 units
3	Why are top predators few?	Little energy reaches the top of the pyramid.
4	An organism that makes its own food	producer
5	Organisms that break down dead matter	decomposers
6	Producers have 50,000 units. Primary consumers?	5,000

#	Question	Answer
7	Secondary consumers?	500
8	Tertiary consumers?	50
9	In grass → rabbit → fox, the fox is a…	secondary consumer
10	If a disease kills most rabbits, predict effects on the grass and the foxes. Explain.	Grass increases (fewer eaters); foxes decrease (less food) or switch prey.
11	Explain the difference between how matter and energy move through an ecosystem.	Matter cycles (decomposers return it); energy flows one way and is lost as heat.

LESSON 1.8 · TEACHER PLAN
Week 4 · Lesson 2 of 2
Unit 1: Cycles of Matter and Energy

Lesson 1.8: Biodiversity and Ecosystem Services

Student goal	I can explain why biodiversity matters and evaluate ways to protect ecosystem services.
Objective	Students explain biodiversity and evaluate solutions.
Standard	MS-LS2-5 — Evaluate competing design solutions for maintaining biodiversity and ecosystem services.
Pacing	Unit 1, week 4, lesson 2 of 2
Time	About 50 minutes
Materials	String, sticks, magnifier, workbook Lesson 1.8, Unit 1 Review
Get ready	Read the student lesson, gather and test the investigation materials, and answer the Check Your Understanding questions.

Lesson Flow

Part	Min	What to do and say
Engage	5	No bees, no breakfast?
Investigate	15	Biodiversity Survey.
Explain	10	Teach services; model evaluating solutions with criteria.
Guided Practice	7	Check Your Understanding 1–3.
Independent	10	Workbook; then the Unit 1 Review.
Exit Ticket	3	Next: Unit 1 Test.

COMMON MISCONCEPTIONS

- Thinking biodiversity only means animals.

SUPPORT

- Criteria-and-constraints chart.

EXTEND

- Plant a pollinator pot.

STUDENT LESSON 1.8 · WITH ANSWERS

Lesson 1.8: Biodiversity and Ecosystem Services

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 1.8 in this book with your teacher.
- [ ] Do the Investigate activity: Biodiversity Survey.
- [ ] Answer the Check Your Understanding questions in Workbook Lesson 1.8, Part A.
- [ ] Complete the rest of Workbook Lesson 1.8 (Parts B–D).

I WONDER

What would happen to your breakfast if all the bees disappeared?

Biodiversity is the variety of species, genes, and ecosystems in a place. Ecosystems with high biodiversity are usually more stable: if one species declines, others can fill its role.

Ecosystem service	Example
pollination	Bees, butterflies, and bats pollinate many fruits, nuts, and vegetables.
clean water	Wetlands filter pollutants as water flows through them.
flood control	Wetlands and forests soak up heavy rain like sponges.
healthy soil	Decomposers and earthworms recycle nutrients.
food and materials	Fish, timber, and medicines come from ecosystems.

EXPLAIN IT · EVALUATE A DESIGN SOLUTION

A town wants to reduce flooding. Option A: build a concrete channel. Option B: restore a wetland upstream.

1 Criteria: reduce flooding, cost, protect biodiversity, clean water.

2 A: moves water fast but removes habitat and doesn't filter water.

3 B: soaks up and slows water, filters pollutants, and provides habitat, but needs land.

Explanation: Option B meets more criteria, especially biodiversity and clean water, though land and cost are constraints.

INVESTIGATE · BIODIVERSITY SURVEY

How many kinds of living things live in a small area?

You need: 4 sticks and string to mark a 1 m × 1 m square; magnifying glass; paper and a pencil.

1 Mark a square in a lawn and another in a less-mowed area (garden edge, field, or woods).

2 Count the different kinds of plants and animals in each square for 10 minutes.

3 Compare the two squares.

Record: a table of species counts and an explanation of which area has more biodiversity and why.

★ KEY IDEA

Biodiversity keeps ecosystems stable and provides services people depend on; good design solutions protect it.

✎ CHECK YOUR UNDERSTANDING

Answers are shown in color.

1. What is biodiversity?

Answer: the variety of living things in an area

2. Name two ecosystem services.

Answer: e.g. pollination, clean water, flood control

3. Why is a diverse ecosystem more stable?

Answer: Other species can fill roles if one declines.

Workbook Answer Key · Lesson 1.8

#	Question	Answer
1	What is biodiversity?	the variety of living things in an area
2	Name two ecosystem services.	e.g. pollination, clean water, flood control
3	Why is a diverse ecosystem more stable?	Other species can fill roles if one declines.
4	An animal that carries pollen	pollinator
5	Wetlands help prevent…	flooding
6	Lawn square: 3 species. Field square: 14 species. Which has more biodiversity?	the field square

#	Question	Answer
7	Explain one way the loss of pollinators would affect people.	Fewer fruits, nuts, and vegetables would be produced.
8	Evaluate: A farmer wants to remove all hedgerows to plant more crops. Name one benefit and one cost to ecosystem services.	Benefit: more crop area. Cost: loss of habitat for pollinators and pest-eating birds; more erosion.
9	Choose one ecosystem service and explain how people could protect it.	Answers vary; e.g. plant native flowers to support pollinators.

UNIT REVIEW · ANSWER KEY
Study before the unit test

About 40 minutes

Unit 1 Review — Answer Key

Homework or a study session before the test.

Answers

#	Question	Answer
1	The variable you measure	dependent
2	Plants releasing water vapor	transpiration
3	Changing N_2 into usable forms	nitrogen fixation
4	Burning fuel	combustion
5	Photosynthesis equation (words)	carbon dioxide + water + light → glucose + oxygen
6	Producers 30,000 units; primary consumers?	3,000
7	Why do energy pyramids narrow at the top?	Only about 10 percent of energy passes to each level; the rest is used or lost as heat.

UNIT TEST · ANSWER KEY
At the end of the unit

About 45 minutes

Unit 1 Test — Answer Key

Answers

#	Question	Answer
1	Does the temperature of water affect how much sugar dissolves? Independent variable:	water temperature
2	Dependent variable:	amount of sugar dissolved
3	Name one controlled variable.	e.g. amount of water, stirring, type of sugar
4	Which process adds water vapor to the air from plants? A. condensation B. transpiration C. infiltration D. runoff	B. transpiration
5	What force pulls precipitation down?	gravity
6	Which process removes carbon dioxide from the atmosphere? A. combustion B. decomposition C. photosynthesis D. respiration	C. photosynthesis
7	Why can't animals use nitrogen gas directly?	It must be fixed into compounds (by bacteria) first.
8	Name the organisms that return nutrients from dead matter to soil.	decomposers
9	Write the word equation for photosynthesis.	carbon dioxide + water (+ light) → glucose + oxygen
10	Write the word equation for cellular respiration.	glucose + oxygen → carbon dioxide + water + energy
11	Producers store 40,000 units of energy. About how much reaches secondary consumers?	400 units
12	Which is an ecosystem service? A. a parking lot B. wetlands filtering water C. a concrete channel	B. wetlands filtering water
13	Leaf disks floating after 12 minutes: bright light 10, dim light 5, dark 0. Write a CER explaining the results.	Rubric: claim that more light increases photosynthesis (1); evidence with numbers (1); reasoning that photosynthesis produces oxygen that makes disks float (1); clear connection of light energy to the process (1).

#	Question	Answer
14	Use a food chain (grass → rabbit → fox) to explain how matter cycles and energy flows. Include photosynthesis, respiration, and decomposers.	Rubric: photosynthesis captures light energy in grass (1); energy passes up with ~10 percent at each level and is lost as heat (1); respiration releases energy and CO_2 (1); decomposers recycle matter to soil (1).

Scoring

Score	What it means / next step
17–20 (85%+)	Mastered.
14–16	Proficient; review missed skills.
Below 14	Reteach and retest.

UNIT 2 · TEACHER OVERVIEW

Properties and Changes of Matter

Atoms and molecules, physical properties and states, changes of state, density, solubility and conductivity, chemical properties, and chemical changes.

Everything around you, from the air you breathe to the phone in your hand, is made of tiny atoms. In this unit you'll build models of molecules, identify substances by their properties, explain how particles behave when matter melts or boils, calculate density, separate mixtures, and finally test whether matter can ever disappear in a chemical reaction.

Big ideas

- Model atoms, molecules, and compounds.
- Describe and measure physical properties, including density.
- Explain changes of state in terms of particle motion and energy.
- Distinguish physical and chemical properties and changes using evidence.
- Test conservation of mass in a chemical reaction.

Standards

Code	Standard
MS-PS1-1	Model the atomic composition of simple molecules.
MS-PS1-2	Analyze properties of substances before and after they interact.
MS-PS1-4	Model particle motion and changes of state.
MS-PS1-5	Model conservation of atoms in a reaction.

What students should already know

- Science practices, data tables, and CER (Unit 1).

Materials for the unit

- Safety goggles and gloves
- Kitchen scale and measuring cups
- Thermometer and timer

- Baking soda, vinegar, salt, sugar, cornstarch
- Steel wool or iron nails
- Magnet, coffee filters, funnel
- Balloons and small plastic bottles

Pacing guide

Teach the lessons in order, then give the week's check. Set the days to fit your own calendar.

Week	Order	Lesson	Title
Week 1	Lesson 1 of 2	2.1	Atoms, Elements, and Molecules
Week 1	Lesson 2 of 2	2.2	Physical Properties and States of Matter
Week 1	End of week	Check	Weekly Check 1: Atoms and Properties
Week 2	Lesson 1 of 2	2.3	Particles in Motion: Changes of State
Week 2	Lesson 2 of 2	2.4	Density
Week 2	End of week	Check	Weekly Check 2: State Changes and Density
Week 3	Lesson 1 of 2	2.5	Solubility, Conductivity, and Mixtures
Week 3	Lesson 2 of 2	2.6	Chemical Properties
Week 3	End of week	Check	Weekly Check 3: Mixtures and Chemical Properties
Week 4	Lesson 1 of 2	2.7	Physical and Chemical Changes
Week 4	Lesson 2 of 2	2.8	Lab: Evidence of a Chemical Reaction
Week 4	Before the test	Review	Unit 2 Review
Week 4	End of unit	Test	Unit 2 Test

TEACHING THIS UNIT AT HOME

Cook something together and look for physical and chemical changes: melting butter is physical; browning toast or baking bread is chemical.

UNIT VOCABULARY

atom	The smallest particle of an element.
compound	Different elements chemically joined.
density	Mass per volume.
thermal energy	The energy of moving particles.
chemical change	A change that makes new substances.
conservation of mass	Mass is not created or destroyed in a reaction.

LESSON 2.1 · TEACHER PLAN
Week 1 · Lesson 1 of 2
Unit 2: Properties and Changes of Matter

Lesson 2.1: Atoms, Elements, and Molecules

Student goal	I can describe atoms, elements, molecules, and compounds and model simple molecules.
Objective	Students model atoms, molecules, and compounds.
Standard	MS-PS1-1 — Develop models to describe the atomic composition of simple molecules (MS-PS1-1).
Pacing	Unit 2, week 1, lesson 1 of 2
Time	About 50 minutes
Materials	Marshmallows/gumdrops or clay, toothpicks, workbook Lesson 2.1
Get ready	Read the student lesson, gather and test the investigation materials, and answer the Check Your Understanding questions.

Lesson Flow

Part	Min	What to do and say
Engage	5	Cutting foil forever.
Investigate	15	Model Molecules.
Explain	10	Read formulas together. Compare element and compound properties.
Guided Practice	7	Check Your Understanding 1–3.
Independent	10	Workbook lesson pages: Warm-Up through Exit Ticket.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Applying a subscript to the wrong atom.

SUPPORT

- Formula-reading flowchart.

EXTEND

- Build a model of ethanol, C_2H_5OH.

STUDENT LESSON 2.1 · WITH ANSWERS

Lesson 2.1: Atoms, Elements, and Molecules

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 2.1 in this book with your teacher.
- [ ] Do the Investigate activity: Model Molecules.
- [ ] Answer the Check Your Understanding questions in Workbook Lesson 2.1, Part A.
- [ ] Complete the rest of Workbook Lesson 2.1 (Parts B–E).

I WONDER

If you cut a piece of aluminum foil in half, then in half again, over and over, would you ever reach a piece you couldn't cut?

All matter, from air to rocks to you, is made of tiny particles called atoms. Atoms are so small that a single grain of sand contains more than a billion billion of them. An element is a pure substance made of only one kind of atom. Scientists know of 118 elements, organized on the periodic table. Each has a one- or two-letter symbol, such as O for oxygen, C for carbon, and Fe for iron.

Atoms can join to form molecules. An oxygen molecule (O_2) is two oxygen atoms. When atoms of different elements join in a fixed ratio, they form a compound. Water (H_2O) is always two hydrogen atoms joined to one oxygen atom. A compound has properties very different from its elements: hydrogen and oxygen are gases, but together they make liquid water.

Substance	Formula	Atoms in one molecule	Element or compound?
oxygen gas	O_2	2 oxygen	element
water	H_2O	2 hydrogen, 1 oxygen	compound
carbon dioxide	CO_2	1 carbon, 2 oxygen	compound
table sugar (sucrose)	$C_{12}H_{22}O_{11}$	12 carbon, 22 hydrogen, 11 oxygen	compound

EXPLAIN IT · READ A CHEMICAL FORMULA

How many atoms are in one molecule of ammonia, NH_3?

1 N stands for nitrogen. With no small number after it, there is 1 nitrogen atom.

2 H stands for hydrogen. The subscript 3 means 3 hydrogen atoms.

3 Add them: 1 + 3 = 4.

Explanation: One molecule of NH_3 has 4 atoms: 1 nitrogen and 3 hydrogen.

INVESTIGATE · MODEL MOLECULES

How can models show the structure of molecules?

You need: mini marshmallows and gumdrops (or clay in two or three colors); toothpicks.

1 Choose a color for each element: for example, white = hydrogen, red = oxygen, black = carbon.

2 Build H_2O, O_2, and CO_2 using toothpicks as bonds.

3 Build CH_4 (methane): one carbon in the center with four hydrogens.

4 Sketch each model and write its formula.

Record: labeled sketches of four molecule models with a color key.

! WATCH OUT

The subscript belongs to the atom just before it. In CO_2, the 2 means two oxygen atoms, not two carbon atoms.

★ KEY IDEA

All matter is made of atoms. Elements contain one kind of atom; molecules are atoms joined together; compounds join different elements in a fixed ratio and have new properties.

✎ CHECK YOUR UNDERSTANDING

Answers are shown in color.

1. What is the difference between an element and a compound?

Answer: An element has one kind of atom; a compound has two or more different elements joined in a set ratio.

2. How many atoms are in one molecule of CO_2?

Answer: 3 (1 carbon and 2 oxygen)

3. Is O_2 an element or a compound?

Answer: an element (only oxygen atoms)

Workbook Answer Key · Lesson 2.1

#	Question	Answer
1	What is the difference between an element and a compound?	An element has one kind of atom; a compound has two or more different elements joined in a set ratio.
2	How many atoms are in one molecule of CO_2?	3 (1 carbon and 2 oxygen)
3	Is O_2 an element or a compound?	an element (only oxygen atoms)
4	Anything with mass that takes up space	matter
5	Two or more atoms joined together	molecule
6	H_2O_2 (hydrogen peroxide)	4 atoms: 2 H, 2 O
7	NaCl (table salt)	2 atoms: 1 Na, 1 Cl
8	$C_6H_{12}O_6$ (glucose)	24 atoms: 6 C, 12 H, 6 O
9	He (helium)	1 atom
10	N_2	element
11	CO	compound
12	Fe	element
13	CH_4	compound
14	Why does water have different properties from hydrogen and oxygen?	When elements combine into a compound, the new substance has its own properties.

LESSON 2.2 · TEACHER PLAN
Week 1 · Lesson 2 of 2
Unit 2: Properties and Changes of Matter

Lesson 2.2: Physical Properties and States of Matter

Student goal	I can describe physical properties and compare solids, liquids, and gases.
Objective	Students describe physical properties and states of matter.
Standard	MS-PS1-2, MS-PS1-4 — Analyze and interpret data on properties of substances (MS-PS1-2).
Pacing	Unit 2, week 1, lesson 2 of 2
Time	About 50 minutes
Materials	Powders, magnifiers, water, goggles, workbook Lesson 2.2
Get ready	Read the student lesson, gather and test the investigation materials, and answer the Check Your Understanding questions.

Lesson Flow

Part	Min	What to do and say
Engage	5	Salt vs sugar.
Investigate	15	Property Detective.
Explain	10	Chart the three states with particle sketches.
Guided Practice	7	Check Your Understanding 1–3.
Independent	10	Workbook. The Weekly Check comes next.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Thinking gases have no mass.

SUPPORT

- Particle-arrangement cards.

EXTEND

- Find a fourth state of matter (plasma) and where it occurs.

STUDENT LESSON 2.2 · WITH ANSWERS

Lesson 2.2: Physical Properties and States of Matter

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 2.2 in this book with your teacher.
- [ ] Do the Investigate activity: Property Detective.
- [ ] Answer the Check Your Understanding questions in Workbook Lesson 2.2, Part A.
- [ ] Complete the rest of Workbook Lesson 2.2 (Parts B–E).

I WONDER

How could you tell the difference between salt and sugar without tasting them?

A physical property can be observed or measured without changing the substance into something new. Color, texture, hardness, odor, mass, volume, melting point, boiling point, and whether a metal is malleable are all physical properties. Scientists use them to identify substances.

The most familiar physical property is state. Matter on Earth is usually a solid, liquid, or gas. The difference is in how the particles are arranged and how they move.

	Solid	Liquid	Gas
shape	definite	takes the shape of its container	fills its container
volume	definite	definite	changes to fill the container
particles	packed tightly; vibrate in place	close together; slide past each other	far apart; move fast in all directions
example	ice	water	water vapor (steam)

EXPLAIN IT · IDENTIFY SUBSTANCES BY PROPERTIES

Two white powders: A dissolves easily and its crystals are cubes; B melts and turns brown when heated in a pan.

1 Cube-shaped crystals are a property of table salt.

2 Melting and browning when heated is a property of sugar.

3 Properties, not appearance alone, identify each one.

Explanation: A is salt; B is sugar.

INVESTIGATE · PROPERTY DETECTIVE

Which properties can identify an unknown substance?

You need: small samples of salt, sugar, baking soda, and cornstarch; a magnifying glass; cups of water; vinegar; a data table.

1 Observe each powder with the magnifying glass and describe its crystals.

2 Rub a pinch between your fingers and describe its texture.

3 Stir a spoonful into water: does it dissolve?

4 Have someone give you a mystery sample and identify it from your table.

Record: a properties table and the identity of the mystery sample, with your evidence.

Safety: Don't taste any samples. Wear safety goggles.

★ KEY IDEA

Physical properties can be observed without changing a substance. Solids, liquids, and gases differ in shape, volume, and how their particles move.

✎ CHECK YOUR UNDERSTANDING

Answers are shown in color.

1. Name three physical properties.

Answer: e.g. color, texture, hardness, mass, volume, melting point, state

2. Which state has a definite volume but no definite shape?

Answer: liquid

3. How do gas particles move?

Answer: fast, far apart, in all directions

Workbook Answer Key · Lesson 2.2

#	Question	Answer
1	Name three physical properties.	e.g. color, texture, hardness, mass, volume, melting point, state
2	Which state has a definite volume but no definite shape?	liquid
3	How do gas particles move?	fast, far apart, in all directions
4	The amount of matter in an object	mass
5	The amount of space something takes up	volume
6	Particles vibrate in place	solid

#	Question	Answer
7	Takes the shape of its container but keeps its volume	liquid
8	Fills any container	gas
9	Particles slide past each other	liquid
10	Which properties from your Property Detective table were most useful for identifying the mystery sample? Explain.	Answers vary; e.g. crystal shape and whether it dissolved.
11	Is boiling point a physical property? Explain.	Yes; measuring it doesn't change the substance into a new one (steam is still water).

WEEKLY CHECK · ANSWER KEY
After Lesson 2.2

About 15 minutes

Weekly Check 1: Atoms and Properties — Answer Key

Answers

#	Question	Answer
1	A pure substance made of one kind of atom	an element
2	Number of atoms in H_2O	3
3	Element or compound: CO_2	compound
4	State with definite volume but no definite shape	liquid
5	Physical property: name one	e.g. color, mass, melting point, hardness
6	Describe how particles move in a solid and in a gas.	Solid: packed tightly, vibrating in place. Gas: far apart, moving fast in all directions.

LESSON 2.3 · TEACHER PLAN
Week 2 · Lesson 1 of 2
Unit 2: Properties and Changes of Matter

Lesson 2.3: Particles in Motion: Changes of State

Student goal	I can explain how adding or removing thermal energy changes particle motion, temperature, and state.
Objective	Students relate thermal energy, particle motion, and changes of state.
Standard	MS-PS1-4 — Develop a model that predicts changes in particle motion, temperature, and state when thermal energy is added or removed (MS-PS1-4).
Pacing	Unit 2, week 2, lesson 1 of 2
Time	About 50 minutes
Materials	Crushed ice, thermometers, timer, workbook Lesson 2.3
Get ready	Read the student lesson, gather and test the investigation materials, and answer the Check Your Understanding questions.

Lesson Flow

Part	Min	What to do and say
Engage	5	Disappearing puddle.
Investigate	15	Ice Melt Data.
Explain	10	Model particles with students moving faster or slower. Interpret a heating curve.
Guided Practice	7	Check Your Understanding 1–3.
Independent	10	Workbook lesson pages: Warm-Up through Exit Ticket.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Thinking temperature always rises when heat is added.

SUPPORT

- Heating-curve labeling practice.

EXTEND

- Research why salt is spread on icy roads.

STUDENT LESSON 2.3 · WITH ANSWERS

Lesson 2.3: Particles in Motion: Changes of State

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 2.3 in this book with your teacher.
- [ ] Do the Investigate activity: Ice Melt Data.
- [ ] Answer the Check Your Understanding questions in Workbook Lesson 2.3, Part A.
- [ ] Complete the rest of Workbook Lesson 2.3 (Parts B–E).

I WONDER

Why does a puddle disappear on a sunny day, even though it never boils?

Particles of matter are always moving. Adding thermal energy makes them move faster; removing it slows them down. Temperature measures the average motion energy of the particles.

When enough energy is added, particles move fast enough to break free of their arrangement, and the state changes. While a substance is changing state, its temperature stays the same: the added energy goes into separating the particles, not speeding them up. That's why a glass of ice water stays near 0°C until all the ice melts.

Change	From → to	Energy	Example
melting	solid → liquid	added	an ice pop dripping
freezing	liquid → solid	removed	water becoming ice cubes
evaporation / boiling	liquid → gas	added	a puddle drying; a pot boiling
condensation	gas → liquid	removed	drops on a cold glass
sublimation	solid → gas	added	dry ice making fog

EXPLAIN IT · READ A HEATING CURVE

Ice at −10°C is heated steadily. The temperature rises to 0°C, stays flat for 4 minutes, then rises again to 100°C and stays flat.

1 Rising sections: particles speed up and temperature increases.

2 First flat section at 0°C: melting; energy separates particles.

3 Second flat section at 100°C: boiling.

Explanation: Flat parts of a heating curve show changes of state.

INVESTIGATE · ICE MELT DATA

What happens to temperature while ice melts?

You need: a cup of crushed ice; a thermometer; a timer; a stirring spoon.

1 Measure the starting temperature of the ice.

2 Let it sit at room temperature; stir and record the temperature every 2 minutes.

3 Note the time when the ice is half melted and fully melted.

4 Keep recording for 10 more minutes.

Record: a data table and a line graph of temperature over time, with the melting section labeled.

★ KEY IDEA

Adding thermal energy speeds particles up; removing it slows them down. During a change of state, temperature stays constant while particles rearrange.

✎ CHECK YOUR UNDERSTANDING

Answers are shown in color.

1. What happens to particle motion when thermal energy is added?

Answer: Particles move faster.

2. Name the change from gas to liquid.

Answer: condensation

3. Why does the temperature stay at 0°C while ice melts?

Answer: The energy goes into breaking the solid's arrangement, not into speeding up particles.

Workbook Answer Key · Lesson 2.3

#	Question	Answer
1	What happens to particle motion when thermal energy is added?	Particles move faster.
2	Name the change from gas to liquid.	condensation
3	Why does the temperature stay at 0°C while ice melts?	The energy goes into breaking the solid's arrangement, not into speeding up particles.
4	Solid directly to gas	sublimation
5	Liquid to gas at the surface	evaporation
6	a mirror fogging after a shower	condensation; energy removed

#	Question	Answer
7	butter softening in a pan	melting; energy added
8	a lake freezing	freezing; energy removed
9	frost disappearing on a cold, sunny morning without melting	sublimation; energy added
10	Describe your Ice Melt graph. What happened to the temperature while the ice melted, and why?	It stayed near 0°C while melting, then rose; energy went into melting first.
11	Explain, in terms of particles, why a puddle evaporates on a sunny day.	Sunlight adds thermal energy; the fastest particles at the surface escape as gas.

LESSON 2.4 · TEACHER PLAN
Week 2 · Lesson 2 of 2
Unit 2: Properties and Changes of Matter

Lesson 2.4: Density

Student goal	I can calculate density and use it to predict whether an object will sink or float.
Objective	Students calculate density and predict sinking and floating.
Standard	MS-PS1-2 — Analyze and interpret data on the properties of substances (MS-PS1-2).
Pacing	Unit 2, week 2, lesson 2 of 2
Time	About 50 minutes
Materials	Tall glass, honey, soap, oil, small objects, graduated cylinder, scale, workbook Lesson 2.4
Get ready	Read the student lesson, gather and test the investigation materials, and answer the Check Your Understanding questions.

Lesson Flow

Part	Min	What to do and say
Engage	5	Ship vs nail.
Investigate	15	Density Column.
Explain	10	Model D = m ÷ V. Demonstrate displacement.
Guided Practice	7	Check Your Understanding 1–3.
Independent	10	Workbook. The Weekly Check comes next.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Confusing density with weight.

SUPPORT

- Formula triangle for m, D, and V.

EXTEND

- Find the density of a coin using displacement.

STUDENT LESSON 2.4 · WITH ANSWERS

Lesson 2.4: Density

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 2.4 in this book with your teacher.
- [ ] Do the Investigate activity: Density Column.
- [ ] Answer the Check Your Understanding questions in Workbook Lesson 2.4, Part A.
- [ ] Complete the rest of Workbook Lesson 2.4 (Parts B–E).

I WONDER

A huge steel ship floats, but a small steel nail sinks. How can that be?

Density tells how much mass is packed into a certain volume. It's a physical property, and every pure substance has its own density, so density can help identify a material.

Density = mass ÷ volume, or D = m ÷ V. Mass is measured in grams (g) and volume in cubic centimeters (cm^3) or milliliters (mL), so density is in g/cm^3 or g/mL. To find the volume of an irregular object, use displacement: measure how much the water level rises when you drop it in.

Substance	Density (g/cm^3)	In water, it…
cork	about 0.24	floats
ice	0.92	floats
vegetable oil	about 0.92	floats on water
water	1.00	—
aluminum	2.70	sinks
iron	7.87	sinks
gold	19.3	sinks

An object less dense than water floats; one more dense sinks. A steel ship floats because its hollow shape is full of air, so the ship's overall density is less than water's.

EXPLAIN IT · CALCULATE DENSITY

A metal block has a mass of 54 g and a volume of 20 cm^3. What is it made of?

1 Write the formula: D = m ÷ V.

2 Substitute: D = 54 g ÷ 20 cm^3.

3 Divide: D = 2.7 g/cm³.

4 Compare with the table: 2.7 g/cm³ matches aluminum.

Explanation: The block is aluminum, with a density of 2.7 g/cm³.

INVESTIGATE · DENSITY COLUMN

Can liquids of different densities stack?

You need: a tall clear glass; honey, dish soap, water with food coloring, vegetable oil; small objects: a grape, a cork, a plastic bead.

1 Slowly pour about 2 cm of each liquid, in this order: honey, dish soap, colored water, oil.

2 Watch whether the layers mix.

3 Gently drop in each small object.

4 Record where each object stops.

Record: a labeled drawing of your column from densest (bottom) to least dense (top), with each object's position.

! WATCH OUT

Density is not the same as weight. A big log weighs more than a small pebble, but the log floats because wood is less dense than water.

★ KEY IDEA

Density = mass ÷ volume. Objects less dense than water float, and denser objects sink. Density helps identify substances.

✎ CHECK YOUR UNDERSTANDING

Answers are shown in color.

1. A rock has a mass of 30 g and a volume of 10 cm³. What is its density?

Answer: 3 g/cm³

2. Will an object with a density of 0.6 g/cm³ float in water?

Answer: yes

3. How can you find the volume of an irregular object?

Answer: by water displacement

Workbook Answer Key · Lesson 2.4

#	Question	Answer
1	A rock has a mass of 30 g and a volume of 10 cm³. What is its density?	3 g/cm³
2	Will an object with a density of 0.6 g/cm³ float in water?	yes
3	How can you find the volume of an irregular object?	by water displacement
4	Mass packed into a volume	density
5	The upward push of a liquid	buoyancy
6	m = 40 g, V = 20 cm³	2 g/cm³
7	m = 79 g, V = 10 cm³	7.9 g/cm³ (iron)
8	m = 12 g, V = 50 cm³	0.24 g/cm³ (cork)
9	m = 193 g, V = 10 cm³	19.3 g/cm³ (gold)
10	A water level rises from 50 mL to 65 mL when a stone is dropped in. The stone's mass is 45 g. Find its volume and density, and predict whether it floats.	V = 15 mL (15 cm³); D = 3 g/cm³; it sinks.
11	Explain why a steel ship floats but a steel nail sinks.	The ship's hollow shape holds air, making its overall density less than water's; the solid nail is denser than water.

WEEKLY CHECK · ANSWER KEY
After Lesson 2.4

About 15 minutes

Weekly Check 2: State Changes and Density — Answer Key

Answers

#	Question	Answer
1	Gas to liquid	condensation
2	Solid directly to gas	sublimation
3	Density formula	D = m ÷ V
4	Density of a 60 g, 20 cm³ block	3 g/cm³
5	Will a 0.8 g/cm³ object float in water?	yes
6	Why does temperature stay the same while ice melts?	Energy goes into breaking the solid's arrangement, not speeding up particles.

LESSON 2.5 · TEACHER PLAN
Week 3 · Lesson 1 of 2
Unit 2: Properties and Changes of Matter

Lesson 2.5: Solubility, Conductivity, and Mixtures

Student goal	I can describe solubility and conductivity and tell mixtures and solutions apart.
Objective	Students describe solubility, conductivity, mixtures, and solutions.
Standard	MS-PS1-2 — Analyze and interpret data on properties of substances (MS-PS1-2).
Pacing	Unit 2, week 3, lesson 1 of 2
Time	About 50 minutes
Materials	Sand, salt, iron filings, magnet, filters, workbook Lesson 2.5
Get ready	Read the student lesson, gather and test the investigation materials, and answer the Check Your Understanding questions.

Lesson Flow

Part	Min	What to do and say
Engage	5	Sugar vs sand.
Investigate	15	Separate a Mixture.
Explain	10	Distinguish mixtures from compounds. Name the property behind each separation.
Guided Practice	7	Check Your Understanding 1–3.
Independent	10	Workbook lesson pages: Warm-Up through Exit Ticket.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Thinking dissolving makes a substance disappear.

SUPPORT

- Separation method matching cards.

EXTEND

- Grow salt or sugar crystals from a solution.

STUDENT LESSON 2.5 · WITH ANSWERS

Lesson 2.5: Solubility, Conductivity, and Mixtures

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 2.5 in this book with your teacher.
- [ ] Do the Investigate activity: Separate a Mixture.
- [ ] Answer the Check Your Understanding questions in Workbook Lesson 2.5, Part A.
- [ ] Complete the rest of Workbook Lesson 2.5 (Parts B–E).

I WONDER

Why can you stir sugar into tea until it disappears, but sand just sinks to the bottom?

A mixture contains two or more substances that are not chemically joined. Each keeps its own properties, and you can separate them by physical means, such as filtering, using a magnet, or evaporating. Trail mix and sand in water are mixtures you can see. A solution is a mixture so evenly blended that it looks like one substance, such as salt water or air.

Solubility describes how much of a substance, the solute, can dissolve in another, the solvent. Sugar and salt are very soluble in water; sand is not. Solubility usually increases with temperature: you can dissolve more sugar in hot tea than in iced tea.

Conductivity describes how well a material lets heat or electricity pass through. Metals such as copper are excellent conductors, which is why electric wires are made of copper. Rubber, plastic, and glass are insulators. Pure water conducts electricity poorly, but salt water conducts it well.

Mixture	Type	How to separate
sand and water	mixture (not a solution)	filter
iron filings and sand	mixture	magnet
salt water	solution	evaporate the water
air	solution of gases	cool it until gases condense (in industry)

INVESTIGATE · SEPARATE A MIXTURE

Can you separate sand, salt, and iron filings using their properties?

You need: a mixture of sand, salt, and iron filings (or paper clips); a magnet in a plastic bag; a coffee filter and funnel; water; a shallow dish.

1 Pass the magnet over the mixture to remove the iron.

2 Stir the rest into warm water so the salt dissolves.

3 Pour it through the filter to catch the sand.

4 Leave the salty water in the dish for a few days to evaporate.

Record: a flowchart of your steps, naming the property used at each step.

Safety: Wear goggles. Keep the iron filings in a bag on the magnet.

★ KEY IDEA

Mixtures keep their parts' properties and can be separated physically. Solubility and conductivity are physical properties that help identify and separate substances.

✎ CHECK YOUR UNDERSTANDING

Answers are shown in color.

1. Is salt water a mixture or a compound?

Answer: a mixture (a solution)

2. How could you separate iron filings from sand?

Answer: with a magnet

3. Does sugar usually dissolve better in hot or cold water?

Answer: hot water

Workbook Answer Key · Lesson 2.5

#	Question	Answer
1	Is salt water a mixture or a compound?	a mixture (a solution)
2	How could you separate iron filings from sand?	with a magnet
3	Does sugar usually dissolve better in hot or cold water?	hot water
4	The substance that dissolves	solute
5	How well a material passes heat or electricity	conductivity
6	lemonade with sugar dissolved	solution; evaporate
7	pebbles in water	mixture; filter or pour off
8	paper clips in rice	mixture; magnet
9	muddy water	mixture; filter or let settle

#	Question	Answer
10	Explain which property you used at each step of Separate a Mixture.	Magnetism for iron; solubility for salt; particle size (filtering) for sand; evaporation for salt.
11	Why are electric wires made of copper and coated in plastic?	Copper conducts electricity well; plastic is an insulator that keeps it safely inside.

LESSON 2.6 · TEACHER PLAN
Week 3 · Lesson 2 of 2
Unit 2: Properties and Changes of Matter

Lesson 2.6: Chemical Properties

Student goal	I can describe chemical properties and explain how they differ from physical properties.
Objective	Students distinguish chemical from physical properties.
Standard	MS-PS1-2 — Analyze data on properties of substances before and after they interact (MS-PS1-2).
Pacing	Unit 2, week 3, lesson 2 of 2
Time	About 50 minutes
Materials	Steel wool or nails, cups, salt, vinegar, goggles, gloves, workbook Lesson 2.6
Get ready	Read the student lesson, gather and test the investigation materials, and answer the Check Your Understanding questions.

Lesson Flow

Part	Min	What to do and say
Engage	5	Log vs pipe.
Investigate	15	Rust Race (observe over five days).
Explain	10	Model the copper example. Discuss safety and material choices.
Guided Practice	7	Check Your Understanding 1–3.
Independent	10	Workbook. The Weekly Check comes next.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Calling any change of appearance a chemical property.

SUPPORT

- Physical vs chemical T-chart.

EXTEND

- Research why the Statue of Liberty is green.

STUDENT LESSON 2.6 · WITH ANSWERS

Lesson 2.6: Chemical Properties

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 2.6 in this book with your teacher.
- [ ] Do the Investigate activity: Rust Race.
- [ ] Answer the Check Your Understanding questions in Workbook Lesson 2.6, Part A.
- [ ] Complete the rest of Workbook Lesson 2.6 (Parts B–E).

I WONDER

A wooden log and a metal pipe are both solid and brown. Which one would you put in a fireplace? How do you know?

A chemical property describes how a substance can change into a new substance. Unlike physical properties, you can only observe a chemical property by changing the substance. Once wood burns, it's no longer wood: it has become ash, smoke, water vapor, and carbon dioxide.

Chemical property	What it describes	Example
flammability	how easily it burns	paper burns easily; water doesn't
reactivity with oxygen	whether it rusts or tarnishes	iron rusts; gold doesn't
reactivity with acids	whether it fizzes or dissolves in acid	baking soda fizzes in vinegar
reactivity with water	whether it reacts in water	sodium metal reacts violently; copper doesn't

Chemical properties help people choose materials safely. Gold and silver are used in jewelry and electronics partly because they don't react easily with oxygen. Iron is strong but rusts, so steel bridges are painted or coated. Flammable liquids, such as gasoline, must be stored away from flames.

EXPLAIN IT · PHYSICAL OR CHEMICAL PROPERTY?

Decide for each: (a) copper is shiny; (b) copper turns green outdoors over time; (c) copper melts at 1,085°C.

1 (a) Shininess can be observed without changing copper: physical.

2 (b) Turning green means copper reacted with air and water to form a new substance: chemical.

3 (c) Melted copper is still copper: physical.

Explanation: (a) physical, (b) chemical, (c) physical.

INVESTIGATE · RUST RACE

Which conditions make iron rust fastest?

You need: four steel wool pads or iron nails; four cups; water, salt water, vinegar, and an empty (dry) cup; labels.

1 Place one pad or nail in each cup: dry, water, salt water, vinegar.

2 Label the cups and predict which will rust fastest.

3 Observe each day for five days.

4 Record color, texture, and amount of rust.

Record: a daily observation table and a claim about which condition caused the most rust.

Safety: Wear goggles and gloves. Wash hands after handling steel wool.

★ KEY IDEA

Chemical properties, such as flammability and reactivity, describe how a substance can change into a new substance. You observe them only by changing the substance.

✎ CHECK YOUR UNDERSTANDING

Answers are shown in color.

1. Physical or chemical property: iron rusts

Answer: chemical

2. Physical or chemical property: ice melts at 0°C

Answer: physical

3. Why is gold used in electronics?

Answer: It doesn't react easily with oxygen, so it doesn't corrode.

Workbook Answer Key · Lesson 2.6

#	Question	Answer
1	Physical or chemical property: iron rusts	chemical
2	Physical or chemical property: ice melts at 0°C	physical
3	Why is gold used in electronics?	It doesn't react easily with oxygen, so it doesn't corrode.
4	How easily a substance burns	flammability

#	Question	Answer
5	A reaction with oxygen	oxidation
6	silver tarnishes black	chemical
7	aluminum is light and silvery	physical
8	gasoline catches fire easily	chemical
9	sugar dissolves in water	physical
10	baking soda fizzes in vinegar	chemical
11	water boils at 100°C	physical
12	Which Rust Race cup rusted fastest? Use your observations as evidence.	Answers vary; typically salt water or vinegar.
13	Explain why you can't observe a chemical property without changing the substance.	A chemical property describes how it changes into a new substance, so you must make that change to see it.

WEEKLY CHECK · ANSWER KEY
After Lesson 2.6

About 15 minutes

Weekly Check 3: Mixtures and Chemical Properties — Answer Key

Answers

#	Question	Answer
1	A mixture so evenly blended it looks like one substance	a solution
2	Separate iron filings from sand with a…	magnet
3	Physical or chemical property: iron rusts	chemical
4	Physical or chemical property: copper conducts electricity	physical
5	How easily a substance burns	flammability
6	Explain the difference between a physical and a chemical property.	A physical property is observed without changing the substance; a chemical property describes how it changes into a new substance.

LESSON 2.7 · TEACHER PLAN
Week 4 · Lesson 1 of 2
Unit 2: Properties and Changes of Matter

Lesson 2.7: Physical and Chemical Changes

Student goal	I can tell physical changes from chemical changes using evidence.
Objective	Students distinguish physical and chemical changes using evidence.
Standard	MS-PS1-2 — Analyze and interpret data to determine if a chemical reaction has occurred (MS-PS1-2).
Pacing	Unit 2, week 4, lesson 1 of 2
Time	About 50 minutes
Materials	Workbook Lesson 2.7
Get ready	Read the student lesson, gather and test the investigation materials, and answer the Check Your Understanding questions.

Lesson Flow

Part	Min	What to do and say
Engage	5	Toast vs cut bread.
Investigate	15	Demonstrate a few quick changes (crumpling paper, a fizzing tablet).
Explain	10	Chart the evidence of chemical change. Discuss the boiling-water exception.
Guided Practice	7	Check Your Understanding 1–3.
Independent	10	Workbook lesson pages: Warm-Up through Exit Ticket.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Thinking every bubble means a chemical change.

SUPPORT

- Evidence checklist card.

EXTEND

- List ten changes in your kitchen and classify each.

STUDENT LESSON 2.7 · WITH ANSWERS

Lesson 2.7: Physical and Chemical Changes

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 2.7 in this book with your teacher.
- [ ] Answer the Check Your Understanding questions in Workbook Lesson 2.7, Part A.
- [ ] Complete the rest of Workbook Lesson 2.7 (Parts B–E).

I WONDER

Is toasting bread the same kind of change as cutting it? How could you prove it?

In a physical change, a substance changes size, shape, or state, but it's still the same substance. Cutting, folding, crushing, melting, freezing, and dissolving are physical changes.

In a chemical change, or chemical reaction, the atoms of the starting substances, the reactants, rearrange to form new substances, the products, with different properties. Burning, rusting, cooking an egg, and baking a cake are chemical changes.

Evidence of a chemical change	Example
gas bubbles form (not from boiling)	baking soda and vinegar fizz
a color change	a banana browning; iron rusting
a temperature change (heat or cold)	a hand warmer heating up; a cold pack chilling
light is given off	a glow stick; a burning candle
a precipitate (solid) forms	a cloudy solid appears when two clear liquids mix
a new odor	milk turning sour

Evidence suggests a chemical change, but you need to check: boiling water bubbles, yet it's a physical change. The best test is to ask whether a new substance with new properties formed.

EXPLAIN IT · ANALYZE THE EVIDENCE

A student mixes two clear liquids. The mixture turns cloudy and a white solid settles to the bottom.

1 Observation: two clear liquids made a solid.

2 Evidence: a precipitate is a new substance with different properties.

3 Conclusion: the atoms rearranged into a new substance.

Explanation: This is a chemical change; the precipitate is a product.

★ KEY IDEA

Physical changes alter size, shape, or state but not identity. Chemical changes produce new substances; evidence includes gas, color change, temperature change, light, precipitates, and new odors.

✎ CHECK YOUR UNDERSTANDING

Answers are shown in color.

1. Physical or chemical change: melting butter

Answer: physical

2. Physical or chemical change: burning a match

Answer: chemical

3. Name two kinds of evidence of a chemical change.

Answer: e.g. gas bubbles, color change, temperature change, light, precipitate, odor

Workbook Answer Key · Lesson 2.7

#	Question	Answer
1	Physical or chemical change: melting butter	physical
2	Physical or chemical change: burning a match	chemical
3	Name two kinds of evidence of a chemical change.	e.g. gas bubbles, color change, temperature change, light, precipitate, odor
4	A new substance made by a reaction	product
5	A solid formed when two liquids react	precipitate
6	chopping wood	physical
7	frying an egg	chemical
8	dissolving sugar in water	physical
9	a nail rusting	chemical
10	water freezing into ice	physical
11	a cake baking	chemical

#	Question	Answer
12	Boiling water forms bubbles. Is boiling a chemical change? Explain using evidence and the definition.	No; the bubbles are water vapor, the same substance in a new state, so no new substance forms.
13	Prove that toasting bread is a chemical change. Name two pieces of evidence.	Color change (brown), new odor, and it can't be changed back; a new substance formed.

LESSON 2.8 · TEACHER PLAN
Week 4 · Lesson 2 of 2
Unit 2: Properties and Changes of Matter

Lesson 2.8: Lab: Evidence of a Chemical Reaction

Student goal	I can plan and carry out a lab, collect evidence of a chemical reaction, and write a claim, evidence, and reasoning.
Objective	Students collect evidence of a reaction and test conservation of mass.
Standard	MS-PS1-2, MS-PS1-5 — Analyze data to determine if a reaction occurred; model conservation of mass (MS-PS1-2, MS-PS1-5).
Pacing	Unit 2, week 4, lesson 2 of 2
Time	About 50 minutes
Materials	Scale, bottles, balloons, baking soda, vinegar, goggles, workbook Lesson 2.8, Unit 2 Review
Get ready	Read the student lesson, gather and test the investigation materials, and answer the Check Your Understanding questions.

Lesson Flow

Part	Min	What to do and say
Engage	5	Where does the mass go?
Investigate	15	Balloon in a Bottle lab.
Explain	10	Model writing a hypothesis. Compare open and sealed results.
Guided Practice	7	Lab write-up.
Independent	10	Finish the lab write-up; then the Unit 2 Review.
Exit Ticket	3	Next: Unit 2 Test.

COMMON MISCONCEPTIONS

- Concluding mass is lost in reactions.

SUPPORT	EXTEND
• Lab report template.	• Repeat with different amounts of baking soda and graph the balloon size.

STUDENT LESSON 2.8 · WITH ANSWERS

Lesson 2.8: Lab: Evidence of a Chemical Reaction

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 2.8 in this book with your teacher.
- [ ] Do the Investigate activity: Balloon in a Bottle.
- [ ] Answer the Check Your Understanding questions in Workbook Lesson 2.8, Part A.
- [ ] Complete the rest of Workbook Lesson 2.8 (Parts B–E).

I WONDER

When baking soda and vinegar fizz, does the mixture lose mass? Where would the mass go?

In this lab, you'll react baking soda (sodium bicarbonate) with vinegar (acetic acid). The reaction produces carbon dioxide gas, water, and sodium acetate. You'll look for evidence of a chemical change and test whether mass is conserved.

Scientists have shown that in any chemical reaction, atoms are rearranged but never created or destroyed. This is the law of conservation of mass. If you trap the gas in a closed container, the total mass before and after the reaction should be the same. In an open container, the gas escapes, so the mass you measure goes down.

Part of the lab	What to write
question	Is mass conserved when baking soda reacts with vinegar?
hypothesis	If… then… because…
variables	independent: open or sealed container; dependent: total mass
data	mass before and after, plus observations
CER	claim, evidence from your data, reasoning about atoms

INVESTIGATE · BALLOON IN A BOTTLE

Is mass conserved in a chemical reaction?

You need: a kitchen scale; a small plastic bottle; a balloon; baking soda (1 teaspoon); vinegar (about 50 mL); a funnel.

1. Pour the vinegar into the bottle. Use the funnel to put the baking soda into the balloon.
2. Stretch the balloon over the bottle opening without spilling the baking soda. Measure the total mass.

3 Lift the balloon so the baking soda falls in. Record your observations.

4 When the fizzing stops, measure the total mass again. Then repeat with the bottle open (no balloon) and compare.

Record: your hypothesis, a data table of masses and observations for both trials, and a CER paragraph.

Safety: Wear goggles. Point the bottle away from faces.

★ KEY IDEA

Gas, temperature change, and other evidence show a chemical reaction. Atoms are rearranged, not created or destroyed, so mass is conserved in a closed system.

✎ CHECK YOUR UNDERSTANDING

Answers are shown in color.

1. What gas forms when baking soda reacts with vinegar?

Answer: carbon dioxide

2. Why does the open container lose mass?

Answer: The carbon dioxide gas escapes into the air.

3. What does conservation of mass mean?

Answer: Matter is not created or destroyed in a reaction; total mass stays the same.

Workbook Answer Key · Lesson 2.8

#	Question	Answer
1	What gas forms when baking soda reacts with vinegar?	carbon dioxide
2	Why does the open container lose mass?	The carbon dioxide gas escapes into the air.
3	What does conservation of mass mean?	Matter is not created or destroyed in a reaction; total mass stays the same.
4	The factor you measure	dependent variable
5	A testable prediction with a reason	hypothesis
6	Write your hypothesis for the sealed-bottle trial in If… then… because… form.	e.g. If the bottle is sealed with a balloon, then the mass will stay the same, because the gas is trapped and atoms are only rearranged.
7	sealed: mass before (g)	Answers vary.

#	Question	Answer
8	sealed: mass after (g)	Answers vary; about the same as before.
9	open: mass before (g)	Answers vary.
10	open: mass after (g)	Answers vary; less than before.
11	List two pieces of evidence that a chemical reaction happened.	e.g. bubbles of gas, the balloon inflated, the bottle felt colder.
12	Write a CER paragraph answering the lab question.	Claim: mass is conserved. Evidence: sealed mass stayed the same; open mass dropped. Reasoning: atoms are rearranged; escaping gas carried mass out of the open bottle.

UNIT REVIEW · ANSWER KEY
Study before the unit test

About 40 minutes

Unit 2 Review — Answer Key

Homework or a study session before the test.

Answers

#	Question	Answer
1	compound	two or more elements chemically joined
2	density	mass per volume
3	sublimation	solid to gas
4	solution	evenly blended mixture
5	chemical property	how a substance can change into a new one
6	precipitate	solid formed from two liquids
7	Density of 45 g in 15 cm^3	3 g/cm^3
8	Atoms in C_2H_6	8
9	Name three pieces of evidence that a chemical change has happened.	e.g. gas, color change, temperature change, light, precipitate, odor

UNIT TEST · ANSWER KEY
At the end of the unit

About 45 minutes

Unit 2 Test — Answer Key

Answers

#	Question	Answer
1	Which is a compound? A. O_2 B. Fe C. H_2O D. He	C. H_2O
2	How many atoms are in one molecule of CH_4?	5
3	What is the difference between an atom and a molecule?	A molecule is two or more atoms joined together.
4	Name the change from liquid to solid.	freezing
5	When thermal energy is added to a substance, its particles… A. slow down B. speed up C. stop moving D. get bigger	B. speed up
6	Find the density: m = 24 g, V = 8 cm^3.	3 g/cm^3
7	Will the object in the last question sink or float in water?	sink
8	Which is a chemical property? A. boiling point B. color C. flammability D. density	C. flammability
9	How could you separate salt from salt water?	evaporate the water
10	Which is a chemical change? A. ice melting B. paper being cut C. wood burning D. sugar dissolving	C. wood burning
11	Name two kinds of evidence of a chemical change.	e.g. gas, color change, temperature change, light, precipitate, odor
12	In a sealed container, what happens to total mass during a reaction?	It stays the same.
13	A student burns a candle on a scale in an open room and finds the mass goes down. Another student says this proves matter was destroyed. Explain whether they are right, using what you know about chemical changes and conservation of mass.	Rubric: identifies burning as a chemical change (1); names gas products such as carbon dioxide and water vapor (1); explains the gases escape into the air (1); states mass is conserved, so matter was not destroyed (1).

Scoring

Score	What it means / next step
14–16 (85%+)	Mastered.

Score	What it means / next step
12–13	Proficient; review missed skills.
Below 12	Reteach and retest.

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INSIDE THIS YEAR

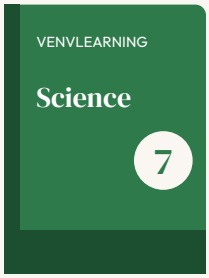
01 Cycles of Matter and Energy

02 Properties and Changes of Matter

IN THIS BOOK

- Week-by-week pacing guide
- Guided lessons with timing and teaching moves
- Misconceptions, supports, and extensions
- Answer keys and scoring guides for every page

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